Summary Notes:
Azure App Services and Backup Solutions

Overview

Here is an overview of the topics covered in this session.

1. Session 1
    a. Azure App Service Plans
        i. Implementing App Service Plans
        ii. Pricing and Features
        iii. Scaling Up and Scaling Out
        iv. Demonstration of App Service Plan Configuration
    b. Azure App Services
        i. Creating and Configuring App Services
        ii. Deployment Slots and Scaling
        iii. Security, Custom Domains, and Backup
        iv. Demonstration of Azure App Services
    c. Azure Container Instances
        i. Containers vs Virtual Machines
        ii. Azure Container Instances and Groups
        iii. Azure Container Apps Overview
        iv. Demonstrations for Azure Containers

2. Session 2
    a. Azure Backup Overview
        i. Services Supported
        ii. Vaults and Policies
        iii. Backup Center Management
    b. Virtual Machine Backups
        i. Snapshots and Recovery Points
        ii. Recovery Services Vault Configuration
        iii. Backup and Restore Operations

c. Azure Site Recovery
    i. Disaster Recovery and Failover
    ii. Recovery from Region Outages

Session 1: Configuring Azure App Services and Containers

1. Azure App Service Plans

- Purpose: Defines compute resources for hosting web apps, API apps, and mobile apps.
- Key Features:
    - Shared Compute: Free or shared plans, no scalability.
    - Dedicated Compute: Basic, Standard, and Premium plans; supports scaling and staging slots.
    - Isolated Plans: Runs on dedicated virtual networks with advanced security.
- Scaling Options:
    - Scale-Up: Increase hardware resources (e.g., CPU, memory).
    - Scale-Out: Add more instances manually or through auto-scaling based on metrics like CPU usage, HTTP requests, or schedules.

2. Azure App Services

- Overview:
    - Fully managed platform-as-a-service (PaaS) offering.
    - Supports .NET, Java, Python, Node.js, and Docker containers.
- Deployment Slots:
    - Allows multiple live environments (e.g., staging and production) for testing before production rollouts.
    - Features like Auto Swap minimize downtime during deployments.
- Security Features:
    - Integration with Azure Key Vault for secrets management.
    - Support for HTTPS with custom SSL certificates.
    - Diagnostic logs for troubleshooting.
- Backup:
    - Supports scheduled and manual backups, with options for selective backups.

3. Azure Container Instances

- Containers vs Virtual Machines:
  - Containers provide lightweight isolation and fast deployments compared to virtual machines.
- Azure Container Instances (ACI):
  - PaaS offering for running containers without managing virtual machines.
  - Supports Linux and Windows containers with fast startup times.
  - Integration with Azure virtual networks for secure communication.
- Azure Container Apps:
  - Provides container orchestration with runtime management (e.g., scaling, failover).
  - Suitable for microservices and job-based workflows.

Session 2: Azure Backup and Disaster Recovery

1. Azure Backup Overview

- What is Azure Backup?:
  - Protects data for Azure VMs, managed disks, file shares, SQL Server, SAP HANA databases, and on-premises resources.
  - Supports built-in backup agents for automated processes.
- Backup Vaults:
  - Two types: Backup vault and Recovery Services vault.
  - Stores backup copies, recovery points, and policies.
- Azure Backup Center:
  - Unified management console for monitoring and configuring backups across environments.
  - Provides policy-based backups for scaling management.

2. Virtual Machine Backups

- Options to Protect Data:
  - Snapshots for quick recovery.
  - Azure Backup for application-consistent backups.
  - Azure Site Recovery for region-level disaster recovery.
- Steps:
  - Create a Recovery Services vault.

        - Configure backup policies defining retention and replication settings.
        - Monitor backups using built-in alerts and logs.
    - Restore Options:
        - Restore entire VMs or individual files from recovery points.
        - Monitor restore jobs through notifications.

3. Azure Site Recovery

- Purpose:
    - Enables disaster recovery by replicating workloads to a secondary region.
    - Supports failover and failback to ensure business continuity during outages.
- Key Features:
    - Continuous replication of workloads.
    - Management of orchestration for recovery plans.
    - Failover to shift operations to secondary regions during disasters.